Heart Disease Risk Modelling

Phase 1: EDA, Bayesian Networks and Classical Baselines

Advanced Machine Learning · Project 15 · UCI Cleveland data

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Prediction exercise only — not clinical decision support

What Phase 1 tests

H1 · Discretisation

- Coarse bins lose information vs continuous inputs
- Same learner, bins vs continuous

H2 · Structure

- Domain graph beats Naive Bayes
- Interaction-capable learner beats additive model

H3 · Probabilities

- Families differ in calibration
- Not only in ranking ability

What the literature says

Models and data

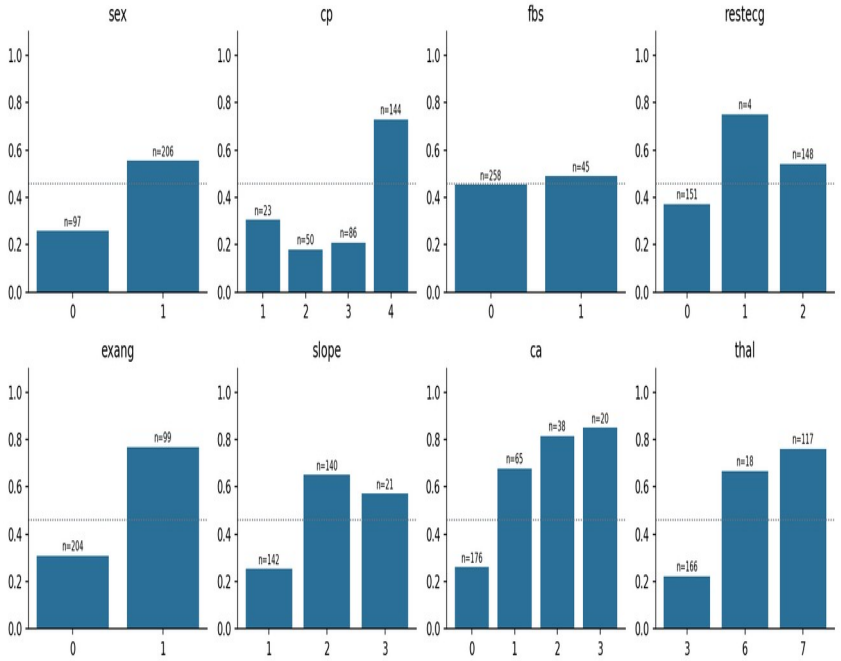
- Cleveland data come from a multi-site coronary disease study (Detrano 1989)
- BN = factorised joint, exact inference (Pearl 1988); BDeu smoothing (Heckerman 1995)
- NB can rank well despite wrong independence (Domingos & Pazzani 1997)
- Richer graphs do not always beat NB (Friedman 1997)

Probability quality and testing

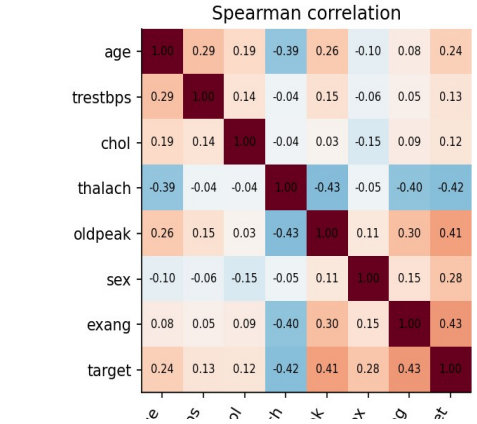
- NB pushes probabilities to 0/1; logistic regression is usually calibrated; forests compress (Niculescu-Mizil & Caruana 2005)
- Neural nets are often over-confident (Guo 2017): check calibration in Phase 2
- Corrected resampled t-test for repeated CV (Nadeau & Bengio 2003)

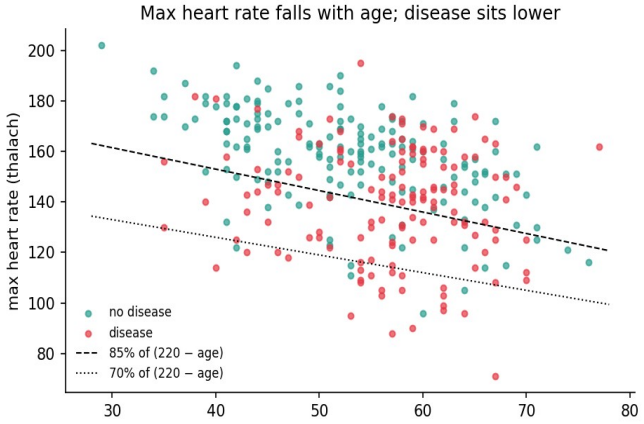
Data and what the EDA shows

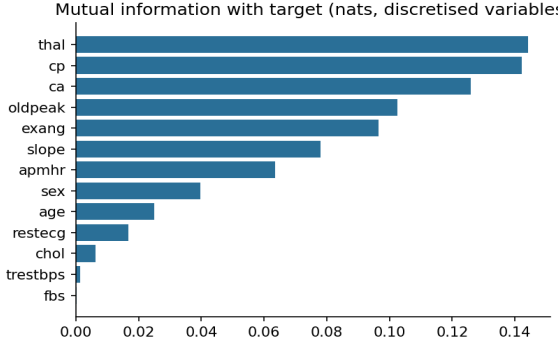
- 303 patients, 13 predictors; target = existing angiographic disease (narrowing > 50%): 139 pos / 164 neg
- 68% male: disease 55% in men vs 26% in women
- Asymptomatic chest pain: 73% disease (n = 144)
- 0 coloured vessels: 26% vs 68–85% with 1–3
- Disease patients: older (56.6 vs 52.6), lower max HR (139 vs 158), more ST depression (1.57 vs 0.59)
- 6 missing values (ca 4, thal 2); no patient IDs, so independence is assumed

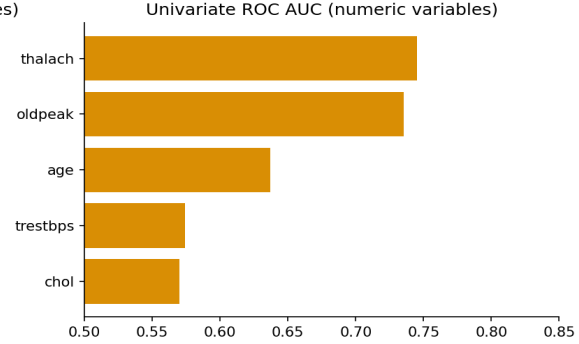


EDA motivates the model structure





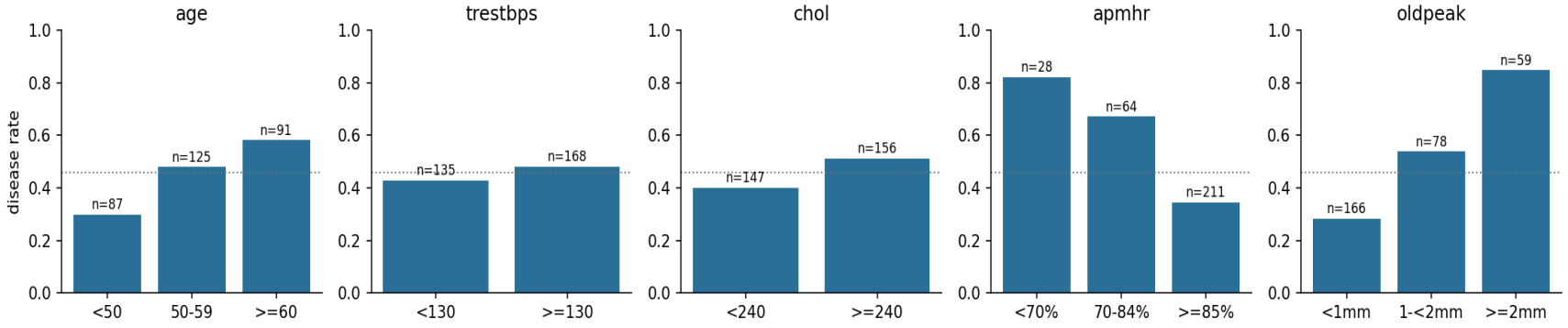




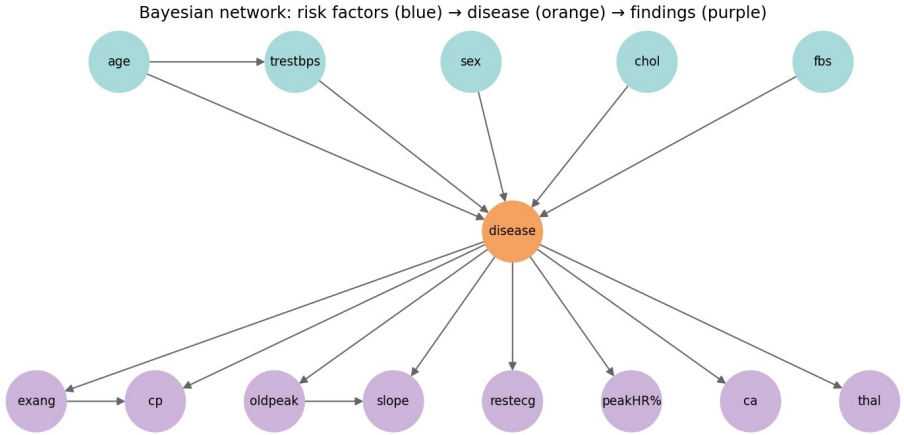
- Max heart rate falls with age (Spearman −0.39)
- Blood pressure rises with age (+0.29) → edge age → BP
- Use peak HR as % of (220 – age)
- Top signals (full-data, exploratory; associations, not causes): thal, cp, ca, oldpeak, exang

Fixed discretisation: modelling choices

Variable	Cut points	Rationale (not a diagnosis)
age	50, 60	decade bands
trestbps	130	common BP threshold
chol	240	common 'high' cholesterol cut
peak HR %	70%, 85% of (220 – age)	age-adjusted HR feature
oldpeak	1, 2 mm	ST-depression bands



Models and fair-comparison protocol



Bayesian net (pgmpy)

14 nodes, 16 edges; risk factors → disease → findings; BDeu (ESS 5); exact inference

Naive Bayes: same pipeline, star graph

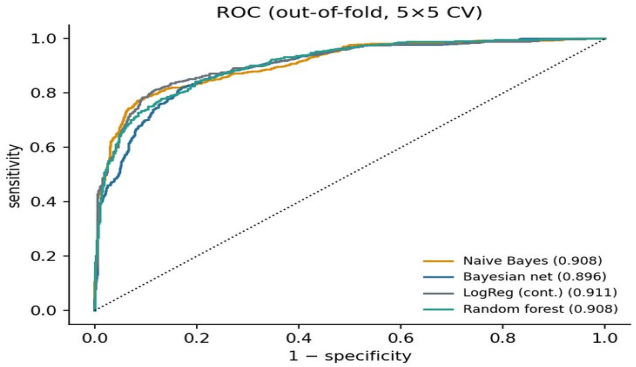
References

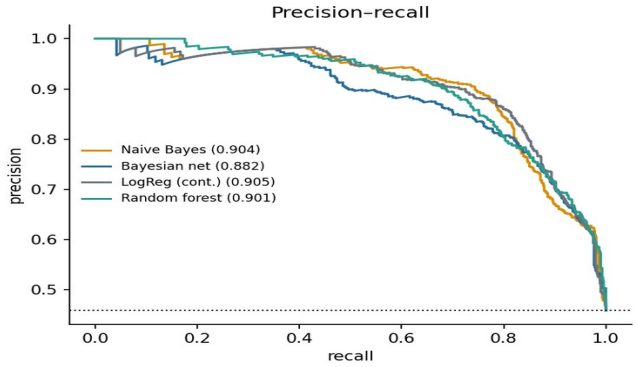
Logistic regression (binned / continuous), random forest, prior-only

Protocol: stratified 5-fold CV × 5 repeats (seed 2026) · same splits for all · training-fold-only preprocessing · ROC AUC, PR AUC (= average precision), sens./spec. at 0.5, Brier, ECE, calibration slope · 500-draw patient bootstrap + corrected t-test (exploratory; not an equivalence test)

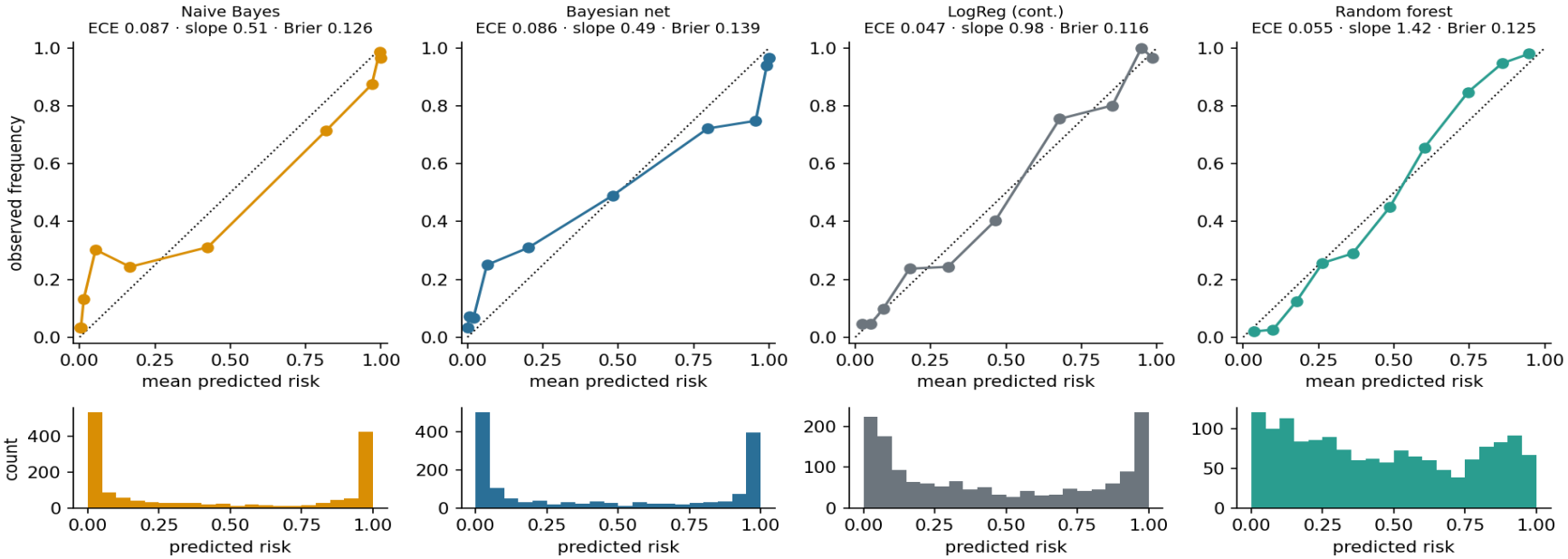
Discrimination: no differences detected

Model	ROC AUC [95% CI]	PR AUC (AP)	Sens.	Spec.	Brier
Naive Bayes	0.908 [0.874, 0.939]	0.904	0.791	0.888	0.126
Bayesian net	0.896 [0.861, 0.927]	0.882	0.778	0.851	0.139
LogReg (disc.)	0.907 [0.870, 0.938]	0.898	0.809	0.888	0.118
LogReg (cont.)	0.911 [0.875, 0.941]	0.905	0.804	0.889	0.116
Random forest	0.908 [0.869, 0.938]	0.901	0.787	0.855	0.125



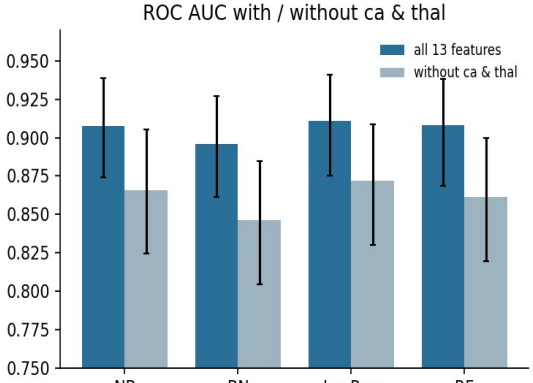


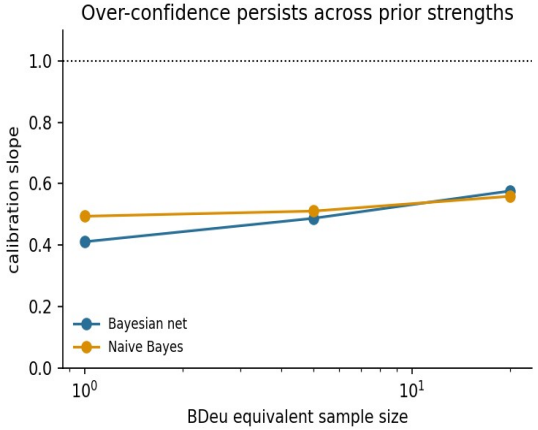
The real difference: calibration



Calibration slope (ideal 1): NB 0.51 · BN 0.49 · LogReg 0.98 · RF 1.42 (below 1 = over-extreme, above 1 = compressed)

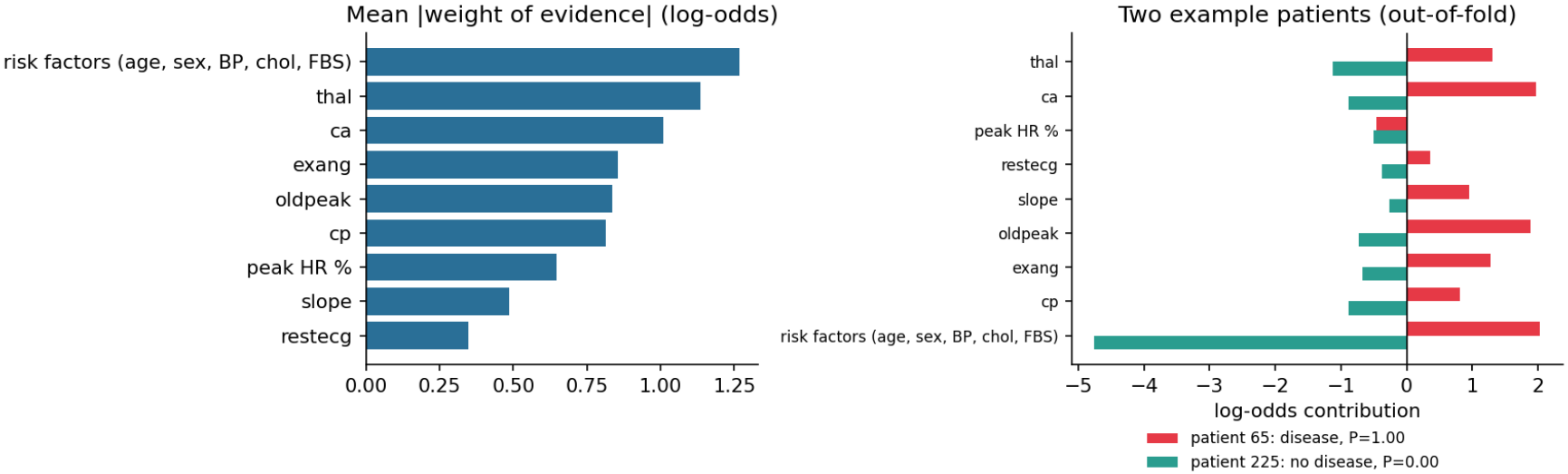
H1 and H2: not supported here





- Matched-HR control: binned − continuous ΔAUC −0.004 [−0.013, 0.004], p = 0.62; no loss detected
- BN not better than NB; RF not better than logistic regression
- On identical binned inputs BN trails logistic regression in Brier (+0.021): points to model form rather than binning
- Without ca/thal: all models −0.04 to −0.05 AUC; BN still over-confident

Exact explanations from the Bayesian net



- Posterior log-odds = risk-factor term + one weight of evidence per finding (exact; checked on the 297 complete-evidence rows)
- Illustration, not a tested cause: in patient 252 several positive findings add up to P = 1.00 (no disease)

Conclusions, limits and Phase 2

Phase 1 conclusions

- Ranking is similar (AUC 0.896–0.911); no differences detected
- No benefit from bins or the domain graph here
- BN and NB over-confident; LogReg near-calibrated
- Clearest gap: probability quality, not ranking

Limits and Phase 2

- n = 303, one clinic, 1980s; no external validation
- EDA-informed graph and cuts; no nested tuning
- Phase 2 (planned): regularised MLP on the same splits, compared on AUC and calibration; SHAP vs BN weights